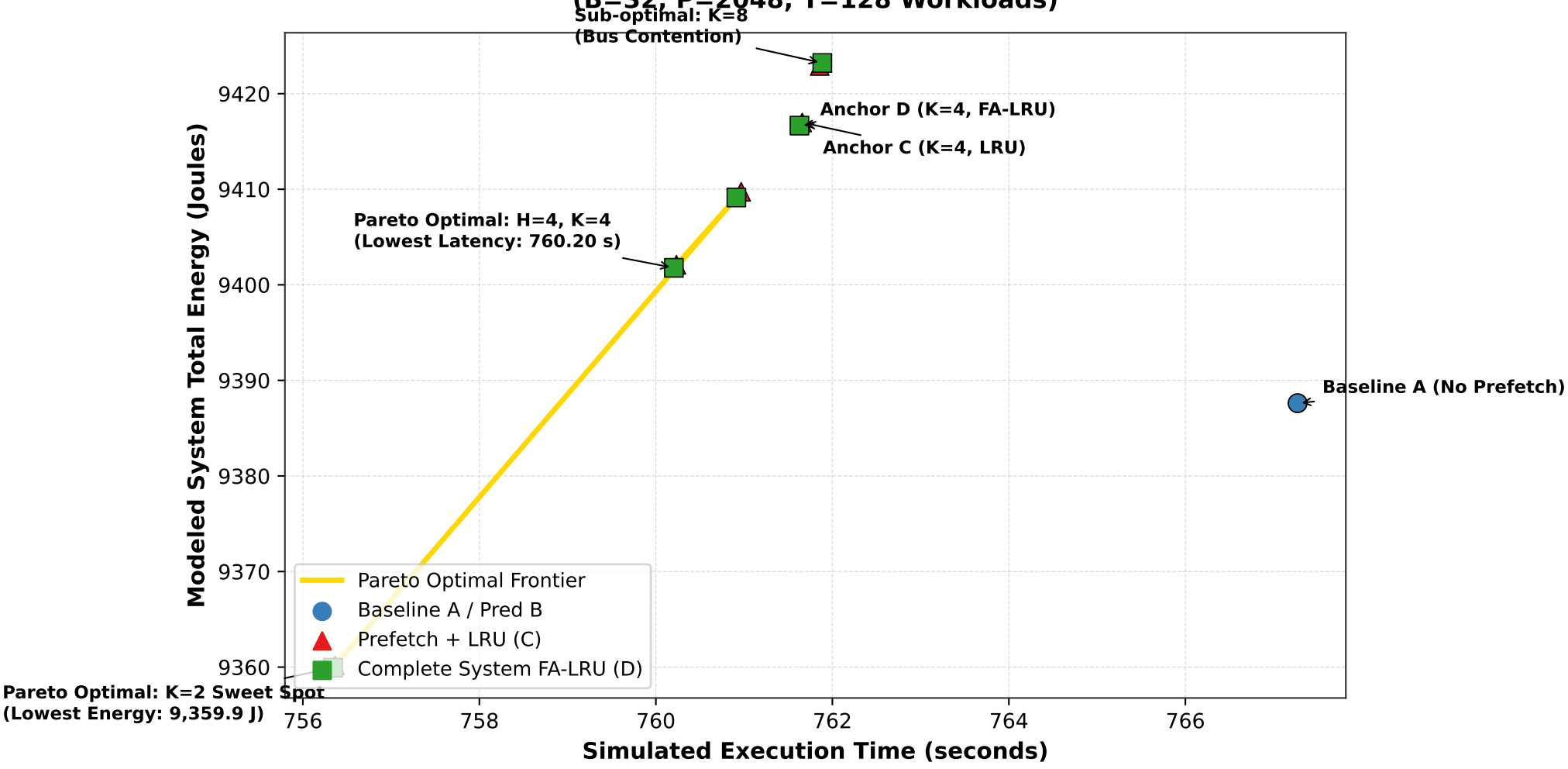
Figure 11: Energy-Performance Trade-off and Pareto Frontier
(B=32, P=2048, T=128 Workloads)
Sub-optimal: K=8
(Bus Contention)
Anchor D (K=4, FA-LRU)
Anchor C (K=4, LRU)
Pareto Optimal: H=4, K=4
(Lowest Latency: 760.20 s)
Baseline A (No Prefetch)
Pareto Optimal: K=2 Sweet Spot
(Lowest Energy: 9,359.9 J)
Pareto Optimal Frontier
Baseline A / Pred B
Prefetch + LRU (C)
Complete System FA-LRU (D)
Modeled System Total Energy (Joules)
Simulated Execution Time (seconds)
9420
9410
9400
9390
9380
9370
9360
756
758
760
762
764
766